

# Md. Forhan Shahriar Fahim

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## EDUCATION

### University of Rajshahi

Jan 2019 – Dec 2024

*B.Sc. in Computer Science & Engineering (CGPA: 3.66, Last 2 Years Avg: 3.82)*

*Rajshahi, Bangladesh*

- **Thesis:** Force Invariant EMG Pattern Recognition (Supervisor: Dr. Shamim Ahmad)
- **Relevant Coursework:** Data Structures (C++), Algorithms, OOP (Java), Database Systems, Operating Systems, AI, Digital Image Processing, Software Engineering, System Analysis.

## EXPERIENCE

### Pundra University of Science & Technology

Mar 2025 – Present

*Lecturer*

*Bogura, Bangladesh*

- Deliver undergraduate courses and develop Outcome-Based Education (OBE)-aligned instructional materials for Algorithms, Simulation & Modeling, Web Engineering, and Structured Programming.
- Supervise undergraduate research theses in Artificial Intelligence (AI) and Machine Learning, guiding students through problem formulation, experimentation, manuscript preparation, and publication.
- Mentor full-stack web development capstone projects, providing technical guidance on system design, implementation, and project management.

## PUBLICATIONS

- [1] S. M. Imran Reza, Iffath Tanjim Moon, **Md. Forhan Shahriar Fahim**, and Md. Tanjid Sheikh, “Explainable Multi-Task Transformer-Based Framework for Cross-Language Source Code Vulnerability Detection with Auxiliary CWE and Language Classification,” *Journal of Cyber Security Technology*, vol. 10, no. 1, Art. no. 2700412, 2026. doi:10.1080/23742917.2026.2700412.
- [2] Md. Tanjid Sheikh, Iffath Tanjim Moon, **Md. Forhan Shahriar Fahim**, Adrita Alam, Sumaya Akter Mahmuda, and S. M. Imran Reza, “Systematic Hybrid Analysis of Android Applications: CWE-Categorized Vulnerabilities and Practical Mitigation Strategies,” in *2026 IEEE 2nd International Conference on Quantum Photonics, Artificial Intelligence & Networking (QPAIN)*, pp. 1–6, 2026. doi:10.1109/QPAIN69676.2026.11546396.
- [3] S. M. Imran Reza, Iffath Tanjim Moon, **Md. Forhan Shahriar Fahim**, Adrita Alam, and Md. Tanjid Sheikh, “An Empirical Analysis of Transformer-Based Models with LIME Explainability for JavaScript Vulnerability Detection,” in *2026 5th International Conference on Electrical, Computer & Telecommunication Engineering (ICECTE)*, pp. 1–6, 2026. doi:10.1109/ICECTE69292.2026.11429428.

## RESEARCH EXPERIENCE

### Systematic Review: SSL in Cancer Imaging | Lead Researcher

Aug 2025 – Present

- Conduct a PRISMA-guided systematic review of Self-Supervised Learning (SSL) methods across MRI, CT, and histopathology for label-efficient medical image analysis.
- Synthesize recent approaches, datasets, evaluation strategies, and research gaps to support the preparation of a manuscript for peer-reviewed publication.

### Benchmarking YOLO Architectures for Crop Disease Segmentation | Lead Researcher Jun 2025 – Present

- Benchmark YOLO architectures (v8–v12) for instance segmentation of crop leaf diseases, comparing predictive performance and computational efficiency across model variants.
- Constructed a custom dataset with pixel-level polygon annotations to support instance-segmentation training and evaluation.
- Evaluate Nano, Small, and Medium model variants and perform hyperparameter tuning to analyze trade-offs between segmentation accuracy and computational cost.

## Force-Invariant EMG Pattern Recognition | Undergraduate Thesis

Apr 2023 – Sep 2024

- Developed a hybrid CNN–LSTM framework for force-invariant sEMG pattern recognition, benchmarking its performance against ResNet-50 and VGG-16 under varying contraction forces.
- Achieved 80% classification accuracy across dynamic contraction levels, demonstrating robustness to variations in muscle contraction force.

## TECHNICAL PROJECTS

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### Biko E-commerce Platform | TypeScript, Express.js, React, MySQL, TypeORM

Mar 2023 – Apr 2024

- Collaborated on a comprehensive e-commerce platform implementing JWT authentication, role-based authorization, and secure transaction flows.
- Implemented optimized CRUD operations for efficient product management, contributing to a scalable and user-friendly online retail solution.
- Developed advanced search and filtering algorithms to improve product discovery and enhance the overall user experience.

### BookHive: Book Store Management | PHP, Laravel, MySQL, Bootstrap

Mar 2023 – Dec 2023

- Architected a robust inventory management system using the Laravel MVC framework to ensure modular code structure and security.
- Implemented comprehensive CRUD functionalities allowing for real-time administration of book records and stock levels.
- Designed a responsive user interface with Bootstrap, integrating dynamic search and filtering logic to streamline database queries.

## ACADEMIC SUPERVISION

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- Undergraduate Research Supervisor (10+ students):** Supervise undergraduate research in medical imaging, including *Brain Tumor Segmentation using Attention Mechanisms* and *Automated Tumor Detection*.
- Capstone Project Supervisor:** Supervise full-stack development projects, including an *Artisanal E-commerce Platform* and a *Web-based Hostel Management System*, with emphasis on software development life cycle (SDLC), system design, implementation, and agile documentation.

## PROBLEM SOLVING

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### Competitive Programming | Codeforces, LeetCode, AtCoder

- Solved 1,000+ algorithmic problems across various platforms; achieved **Max Rating 1396 (Pupil)** on Codeforces and **1661** on LeetCode.

## HONORS & AWARDS

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- Young Enthusiastic Faculty Member Award (2025):** Received from Pundra University of Science & Technology in recognition of dedication to teaching, student mentorship, and departmental engagement.
- Dean's Award (2023):** Recognized for outstanding academic performance during undergraduate studies at the University of Rajshahi.
- Honorable Mention, University of Rajshahi Project Showcase (2019):** Recognized for technical excellence in the university project showcase.
- Education Board Scholarship, SSC (2016):** Awarded the General Grade Scholarship by the Government of Bangladesh for outstanding performance in the Secondary School Certificate examination.
- Talent Pool Scholarship, JSC (2013):** Awarded the highest-tier government scholarship for outstanding performance in the Junior School Certificate examination.

## TECHNICAL SKILLS

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- Programming Languages:** C/C++, Python, Java, PHP.
- Machine Learning & Deep Learning:** PyTorch, TensorFlow.
- Web & Databases:** Node.js, Laravel, MySQL.
- Tools & Platforms:** Git, Linux, LaTeX, Microsoft Office (Word, Excel, PowerPoint).
- Core Concepts:** Object-Oriented Programming (OOP), Computer Networks, Operating Systems, Artificial Intelligence, APIs, Database Normalization, Agile Methodology.

## LEADERSHIP & CO-CURRICULAR ACTIVITIES

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- **Event Coordinator, PUB CSE Fest (2025):** Served on the organizing committee for the annual departmental festival, coordinating sports tournament logistics and supervising the media coverage team.
- **Competitive Programming Mentor:** Mentored students through the PUB and RU programming communities, conducting workshops on algorithms, problem solving, and national programming contest preparation.

## PROFESSIONAL DEVELOPMENT

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- **Integrating Computer Vision in Robotics (Sep 2025):** Seminar on autonomous systems and HRI applications.
- **Research Methodology & Questionnaire Design (Sep 2025):** Faculty training on statistical tools, survey design, and academic best practices.

## LINGUISTIC PROFICIENCY

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- **English:** Professional Working Proficiency
- **Bangla:** Native

## REFERENCES

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### **Dr. Shamim Ahmad**

Professor, Dept. of CSE,  
University of Rajshahi

**Email:** shamim\_cst@ru.ac.bd

*Undergraduate thesis supervisor*

### **Dr. Md. Ekramul Hamid**

Professor, Dept. of CSE,  
University of Rajshahi

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